

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 10

(1) Research Program/Project/Study Title: Smart Agriculture System using Arduino	
Proponent Name:	Cherry Joy A. Palomar
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

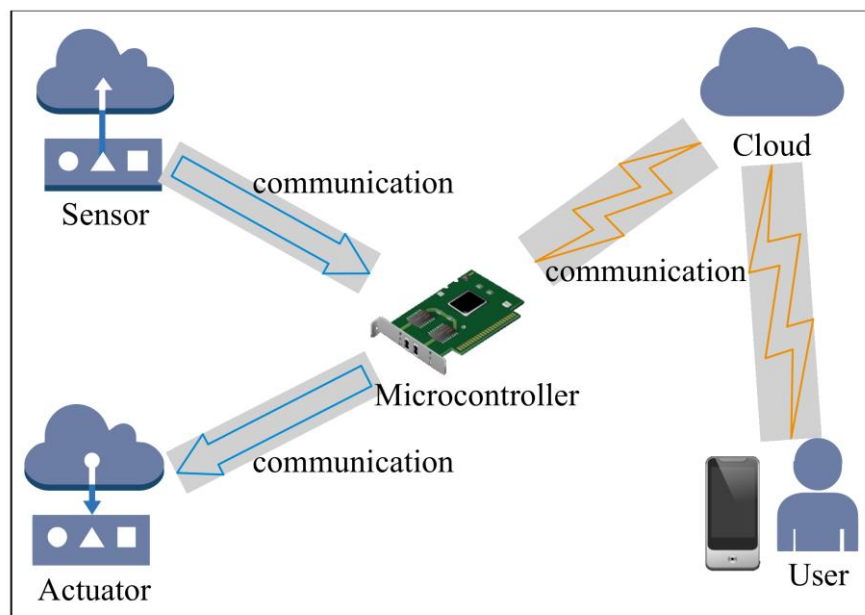
(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:			
Province:			
Region:			
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity 		(8) Discipline/Field of Specialization Information Technology	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 2 of 10

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Agriculture is one of the important businesses that mainly affects the mankind. Farming is the way that human used to harvest plants and consumed them in their daily life. Farming has been improved by many technologies supporting cropping system. The Philippines being an agricultural country need to invest in promoting inclusive growth, and build more sustainable agriculture and food systems, that are resilient to calamities and respond effectively to climate change impact. The main goal is to develop the agriculture sector to be able to attain food self sufficiency, uplift the rural community and increase the income of the farmers. However, agricultural sector has been beset with persistent challenges resulting in low farm incomes, low rural employment, lack of food security, and meager agricultural competitiveness. Through this propose study, a Smart Agriculture System is introduced. The integration of sensors and microcontrollers has revolutionized various fields, including agriculture, by enabling the development of sophisticated plant prediction systems. These systems leverage sensor technology to collect real-time data on environmental factors such as temperature, humidity, soil moisture, and pH level, crucial for plant growth. Coupled with microcontrollers, which process and analyze this data efficiently, these systems empower farmers and growers with actionable insights into what plant is suitable to grow for a specific soil condition. By employing ANN, these systems can determine what possible plants can grow in a specific soil content, based on the data gathered by the sensors. This plant prediction systems empower farmers to become more informed decision-makers, leading to sustainable practices, higher yields, and a more secure future for agriculture. Additionally, such systems streamline the decision-making process by providing tailored recommendations for crop selection, considering soil characteristics such as pH levels, nutrient content, and texture. This personalized approach enhances crop productivity and resilience, optimizing yield outcomes while minimizing input costs and environmental impact. By guiding farmers towards the cultivation of crops best suited to their specific soil conditions, these systems foster sustainable agricultural practices and mitigate the risks associated with unsuitable crop choices. Ultimately, the integration of soil-based predictive models empowers farmers to make informed planting decisions, maximizing profitability, and promoting long-term soil health and productivity.

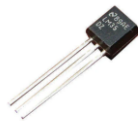
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)



	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 3 of 10



The **Soil Moisture Sensor** uses capacitance to measure the water content of soil (by measuring the dielectric permittivity of the soil, which is a function of the water content). Simply insert this rugged sensor into the soil to be tested, and the volumetric water content of the soil is reported in percent.



A **temperature sensor** is an electronic device that measures the temperature of its environment and converts the input data into electronic data to record, monitor, or signal temperature changes. There are many different types of temperature sensors. Some temperature sensors require direct contact with the physical object that is being monitored (contact temperature sensors), while others indirectly measure the temperature of an object (non-contact temperature sensors).



The **humidity sensor** is a device that senses, measures, and reports the relative humidity (RH) of air or determines the amount of water vapor present in gas mixture (air) or pure gas.



The **ph sensor** is a sensor used to detect the concentration of hydrogen ions in a solution and convert it into a corresponding usable output signal. It is usually composed of a chemical part and a signal transmission part. The measurement range is 0-14 digital representation. The value 7 indicates neutrality.



The **color sensor** is used to measure and to detect the presence of **NPK content of soil**. The color sensor's photodiode is designed to decide the amount of additional contents of these nutrients that has to be added into the soil to increase soil richness and fertility.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 4 of 10



Arduino is an open-source platform (prototype platform) which is based on an easy-to-use hardware as well as software. It comprises a circuit board, which shall be programmed (usually known as a microcontroller) and a software called Arduino IDE (Integrated Development Environment), with which one can write and upload the computer code on to the physical board. The Arduino microcontroller board is very easy to use one yet a powerful single board computer. It has gained considerably a good adhesion in the hobby as well as the professional market. Arduino is referred to as open-source, which means that the hardware is reasonably priced as per the usage and the development software is free of cost.



Smart Agriculture Application

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

- 1) To design Smart Agriculture System that is adaptable to different regions, climates, and soil types.
- 1) To create a system that could determine the moisture, humidity, temperature, pH level and nutrient content of the soil.
- 2) To develop an application that accurately determines the optimal plants to be planted based on the specific characteristics of a given soil.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Crop yield prediction method and system based on soil quality index

Inventors: ZHANG QINGWEN; SHI CHANG

Publication: CN116894514A·2023-10-17

Abstract

The invention relates to the technical field of yield prediction, and particularly provides a crop yield prediction method and system based on a soil quality index, and the method comprises the steps: selecting a soil quality evaluation index, and building a standard scoring function between the soil quality evaluation index and soil quality; screening the soil quality evaluation indexes, establishing a minimum data set evaluation index, and performing soil quality index calculation by using the minimum data set evaluation index; acquiring a real-time growth state of the crop, correcting the soil quality index according to the real-time growth state, and acquiring a corrected soil quality index; and performing yield prediction on a to-be-predicted crop based on the corrected soil quality index. According to the method, accurate soil quality evaluation and crop yield

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 5 of 10

prediction can be provided, effective decision support is provided for agricultural production, farmland management and resource utilization are optimized, and improvement of the efficiency and sustainability of agricultural production is facilitated.

Grain yield prediction system and method based on model

Inventors: Yu Lingxue; Yang Jiuchun

Publication: CN117077868A·2023-11-17

Abstract

The invention discloses a grain yield prediction method based on a model, and relates to the technical field of grain yield prediction, and the method comprises the following steps: collecting related data including historical grain yield data, grain growth condition data, environment data and soil quality data; calculating a crop growth condition evaluation value through the leaf area index, the biomass and the growth rate; obtaining a soil quality evaluation value through the nitrogen content, the phosphorus content and the potassium content of the soil, obtaining an environment evaluation value through the temperature, the humidity and the illumination intensity, and generating a crop growth environment evaluation value; establishing a linear relationship between the crop growth condition evaluation value, the crop growth environment evaluation value and the grain yield by using a regression method, and constructing a prediction model of the grain yield; and when the predicted grain yield does not reach the expected value, analyzing the growth condition evaluation value, the soil quality evaluation value and the environment evaluation value, and taking corresponding measures according to the analysis result.

Soil Management System

Inventors: Kanamaru Yoichi; Yamauchi Eiji; Nakayama Koji; Saito Yoshiaki

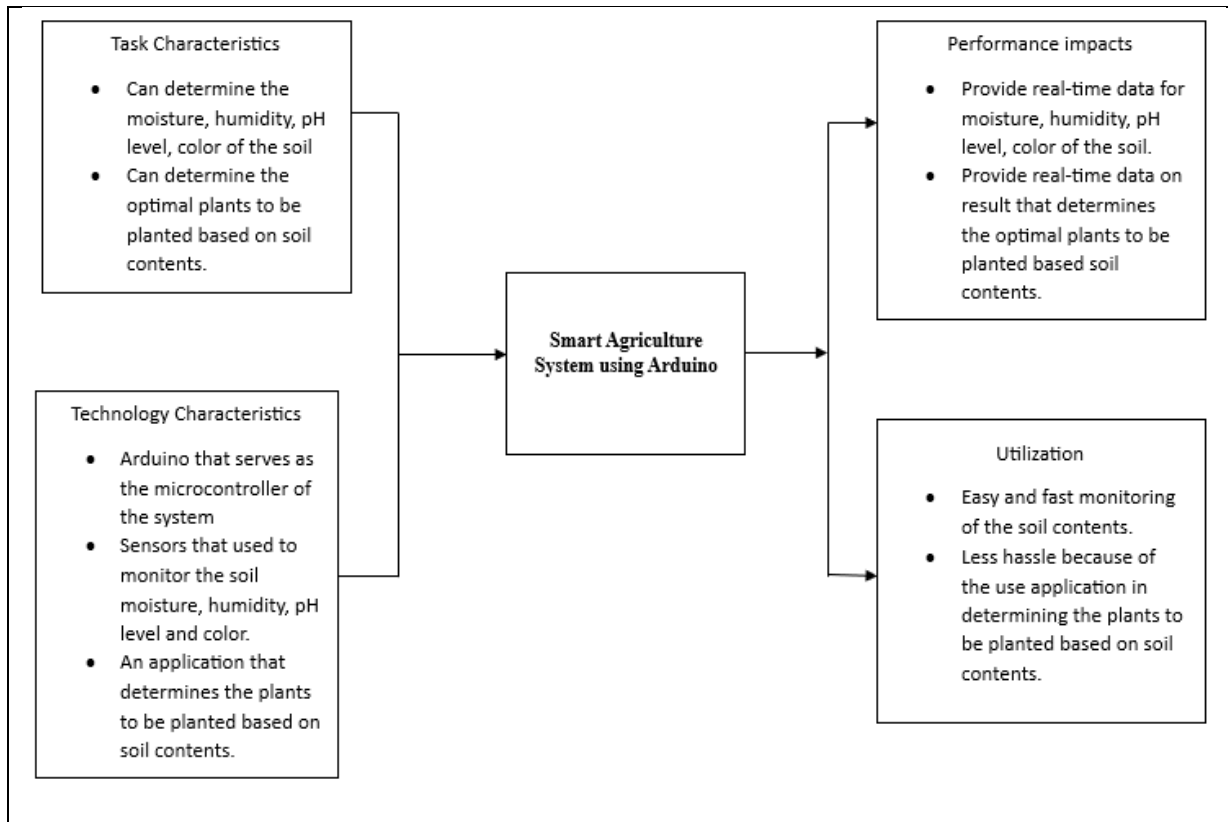
Publication: JP2023157044A·2023-10-26

Abstract

To provide a soil management system prepared for sudden changes in the cultivation environment while reducing power consumption.SOLUTION: A soil management system comprises: a soil sensor 100 equipped with a control unit 110 having a power source, control means, and a switch and a sensor unit 120 having a temperature sensor, a moisture sensor, and a transmitter/receiver section; and a control device 200 equipped with a storage section 240, environmental information acquisition means 302, environmental information prediction means 304, and measurement interval setting means 306. The control means turns on the switch at predetermined measurement intervals to supply power from the power source to the sensor unit. The temperature sensor measures soil temperature. The moisture sensor measures soil moisture content. The environmental information acquisition means acquires the environmental temperature from the soil temperature. The environmental information prediction means predicts temporal changes in dryness based on plant characteristics and environmental information including information on soil temperature, soil moisture content, and environmental temperature. The measurement interval setting means sets a measurement interval based on the predicted degree of dryness and sends the measurement interval to the soil sensor.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

	Document Type: FORM ISO 9001:2015	Document Code	RES-F09
		Revision No.	00
	Document Title: RESEARCH PROPOSAL	Effective Date	January 26, 2023
		Page	Page 6 of 10



(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Waterfall Model - Design

Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project. In "The Waterfall" approach, the whole process of software development is divided into separate phases. In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially.

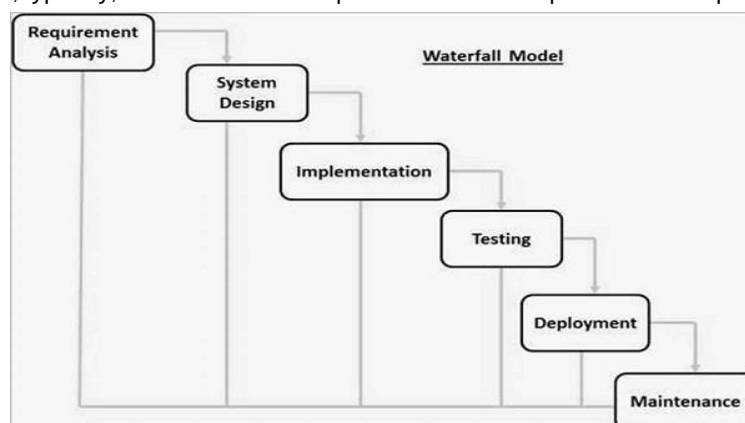
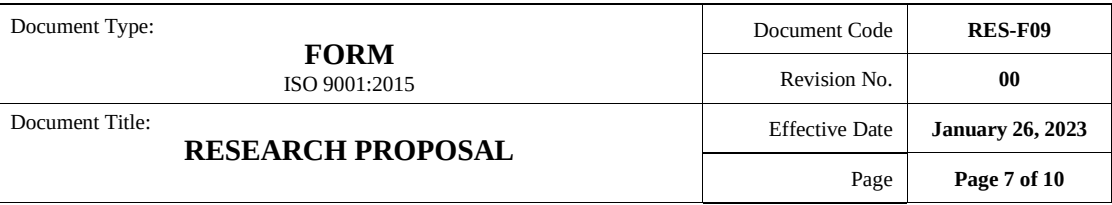


Fig. 6 Waterfall model

The following illustration is a representation of the different phases of the Waterfall Model.

The sequential phases in Waterfall model are –



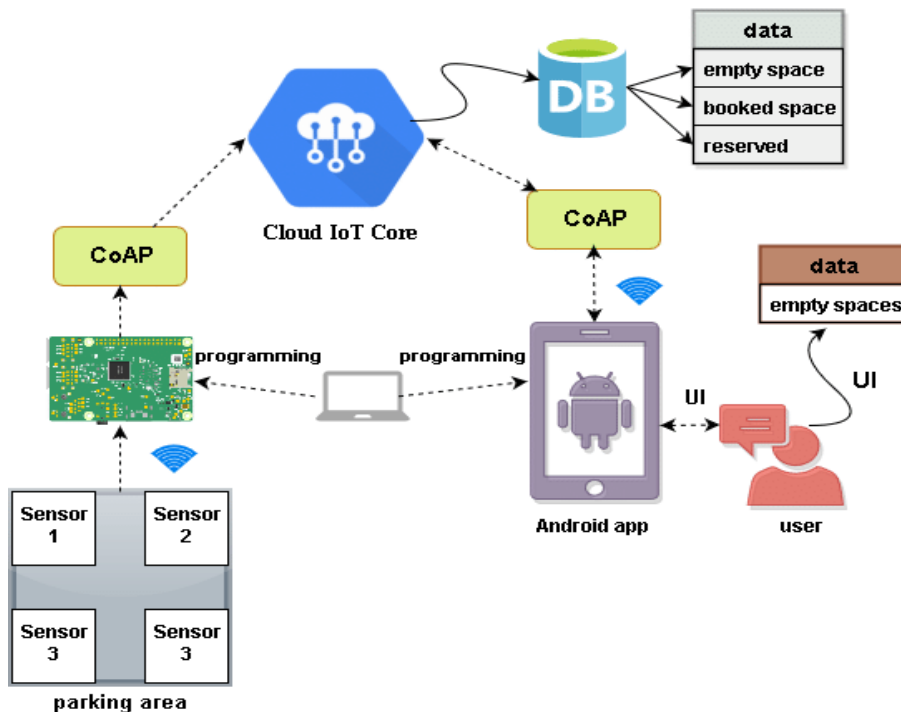
- All these phases are cascaded to each other in which progress is seen as flowing steadily downwards (like a waterfall) through the phases. The next phase is started only after the defined set of goals are achieved for previous phase and it is signed off, so the name "Waterfall Model".

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 8 of 10

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)



(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target user of this system are the farmers. It helps the farmers to have an accurate predictions of optimal plants increase yields and decrease wasted resources, leading to higher profits. This research has the potential to revolutionize agriculture, ensuring food security and environmental well-being, all while putting farmers in control of their future.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
	Programmer (Code)	IT
	Programmer (Arduino)	IT

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 9 of 10

Percent time devoted to the Program/ Project/ Study		
Program/Project Leader (in hours):		
Program/Project Members (in hours):		
(19) Budgetary Requirement		
SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	2	50, 000
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		2, 000
- Meal allowance during data gathering		3, 000
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		1, 000
- Printing		1, 000
C. CAPITAL OUTLAY		
Total		107, 000

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Internet of Things: Architectures, Protocols, and Applications, Pallavi Sethi¹ and Smruti R. Sarangi (2017)
Sensors for daily life: A review, Mohd Javaid a, Abid Haleem a, Shanay Rab a, Ravi Pratap Singh b, Rajiv Suman (2021)
Crop yield prediction method and system based on soil quality index, ZHANG QINGWEN; SHI CHANG
 Publication: CN116894514A·2023-10-17
Grain yield prediction system and method based on model, Yu Lingxue; Yang Jiuchun
 Publication: CN117077868A·2023-11-17
Soil Management System, Kanamaru Yoichi; Yamauchi Eiji; Nakayama Koji; Saito Yoshiaki
 Publication: JP2023157044A·2023-10-26

(21) Capsule Curriculum Vitae (One-page brief C.V. only)

Cherry Joy Arroyo Palomar
Master of Science in Computer Science Graduate

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 10 of 10

PROFESSIONAL EXPERIENCE

August 2021 – Present	College Instructor Computer Studies Department Capiz State University (Mambusao Campus) Poblacion, Mambusao, Capiz, Philippines
June 2013 – July 2021	College Instructor College of Computer Studies Filamer Christian University Roxas City, Philippines

EDUCATION

June 2014 – January 2021	Central Philippine University Master of Science in Computer Science Jaro, Iloilo City
June 2009 – March 2013	West Visayas State University Bachelor of Science in Information Systems Cum Laude
June 2005 – March 2009	Bungsuan National High School Valedictorian Bungsuan, Dumarao, Capiz Philippines
June 1999 – March 2005	Bungsuan Elementary School Valedictorian Bunsuan, Dumarao, Capiz Philippines